

THE PHYSICAL AI DUAL-BRAIN ARCHITECTURE

Privilege Separation & Safety Contracts on Heterogeneous Silicon

UNTRUSTED PROPOSAL ENGINE (Linux MPU / Edge NPU)

UNTRUSTED PROPOSALS

- Multi-Modal Foundation Models (VLMs), Vision Encoders (ViT / DINOv2), Action Chunk Decoders (Diffusion / ACT)
- Asynchronous Deliberation: 1 Hz Semantic Intent $\rightarrow$ 20–50 Hz Action Chunking (Amortizes Inference Delay Across Time)

Emits: Expiring Intent Leases $pt = \langle SE(3) \text{ Target, Workspace Bounding Volume, } t_{\text{expire}}, \text{ Monotonic Counter} \rangle$

Shared SRAM Mailbox · Expiring Proposal pt ($TTL \leq 100 \text{ ms}$)

TRUSTED PERMISSION AUTHORITY (Real-Time Bare-Metal MCU)

SOLE PERMISSION VETO

- Dedicated 1000 Hz Timing Loop: Zero Dynamic Memory Heap (Static SRAM allocation, no malloc / GC jitter)
- Minimal-Intervention Control Barrier Functions (CBF: $h(x) \geq 0$) + Dynamic Stopping Clearance Check ($d_{\text{stop}} \leq d_{\text{clear}}$)

Emergency Interlock (IEC 60204-1): Heartbeat Timeout ($> 20 \text{ ms}$) $\rightarrow$ SS1 Dynamic Braking $\rightarrow$ Safe Torque Off (STO)

Permitted Phase Current Setpoints $ut = \text{permit}(pt)$ (20 kHz PWM)

THE PHYSICAL WORLD (Irreversible State Mutation: $W_t \longrightarrow W_{t+1}$)

NON-REVERSIBLE PHYSICS

- 3-Phase MOSFET Inverter Bridge · Stator Magnetic Flux · Kinetic Momentum ($p = mv$) · Joule Heating (I^2R)
- *Endogenous Feedback: Physical Mutation W_{t+1} Instantly Shapes All Future Sensory Observations O_{t+1}*